S.No. : 438 NBS 4202

No. of Printed Pages : 04

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 49902	Roll No.										

B. Tech. Examination, 2026

(Even Semester)

ENGINEERING PHYSICS

Time : Three Hours] *[Maximum Marks : 60*

Note :– Attempt all questions.

SECTION – A

1. Attempt all parts of the following : $8\times1=8$

   (a) Define interference of light.

   (b) What are matter waves?

   (c) What do you mean by optic axis?

   (d) State Heisenberg's uncertainty principle.

   (e) Why the centre of Newton's rings generally appears dark?

[P. T. O.

(f) What was the objective of Michelson-Morley experiment?

(g) Define equation of continuity.

(h) What are Einstein's postulates?

SECTION–B

2. Attempt any two parts of the following : $2\times6=12$

(a) A sugar solution in a 20 cm long tube produces an optical rotation of 13^{0}. The solution is diluted to one-third of its previous concentration. Find the optical rotation produced by a 30 cm long tube containing the diluted solution.

(b) If the kinetic energy of a body is twice its rest mass energy, find its velocity.

(c) Calculate the acceptance angle, numerical aperture and critical angle for an optical fibre with refractive indices of core and cladding as 1.50 and 1.45 respectively.

(d) Find the energy of an electron moving in one-dimensional infinitely deep potential box of width 1Å.

SECTION–C

Note :– Attempt all questions. Attempt any two parts from each questions. $8\times5=40$

3. (a) Explain the phenomenon of double refraction in Nicol prism. Describe its construction and working.

(b) What are Newton's rings? Prove that in reflected light, the diameter of bright rings are proportional to the square root of odd natural numbers.

(c) Discuss Fraunhofer diffraction due to single slit and show that the relative intensities of central maxima to secondary maxima are in the ratio :

$$1:\frac{4}{9\pi^2}:\frac{4}{25\pi^2}:\frac{4}{49\pi^2}:\ldots\ldots\ldots\ldots$$

4. (a) What is an optical fibre? Differentiate between step index and graded index fibres.

(b) Define phase and group velocity. Derive a relation between them.

(c) Deduce time dependent Schrodinger wave equation.

5. (a) Derive Poynting theorem and give significance of each term.

   (b) Describe the construction and working of Bragg's spectrometer.

   (c) Explain the concept of displacement current. How it led to the modification of Ampere's circuital law?

6. (a) Derive length contraction using Lorentz transformation equations.

   (b) Establish Einstein's mass-energy relation.

   (c) Derive relativistic velocity addition theorem. Show that it is consistent with Einstein's second postulate.

⌘⌘⌘